# Multiple IV Infusions & Multiple Oral Practice 3

Module 6 - Multiple Dosings · Daily practice sheet · No date

Name _____ Date _____

**1a** 1. A 200-mg dose of an antibiotic was administered as an IV infusion over a period of 2 hours. Eight hours after the start of the first infusion, a second dose of 200 mg was infused, again over 2 hours. The antibiotic has a half-life of 4 hours and a total body clearance of 2.8 L/hr. a. What is the plasma drug concentration at the end of the first infusion?

**1b** 1. A 200-mg dose of an antibiotic was administered as an IV infusion over a period of 2 hours. Eight hours after the start of the first infusion, a second dose of 200 mg was infused, again over 2 hours. The antibiotic has a half-life of 4 hours and a total body clearance of 2.8 L/hr. b. What is the plasma drug concentration 4 hours after the cessation of the second infusion?

**2a** 2. A patient is receiving his antihypertensive medication as 500 mg every 8 hours in the form of an oral capsule. The bioavailability of the drug from this capsule is 80% and the total body clearance in this patient is 2.5 L/hr. a. What is the average steady-state plasma concentration of this drug in this patient?

**2b** 2. A patient is receiving his antihypertensive medication as 500 mg every 8 hours in the form of an oral capsule. The bioavailability of the drug from this capsule is 80% and the total body clearance in this patient is 2.5 L/hr. b. What will be the average steady-state plasma concentration if the patient were taking 1000 mg every 8 hours of the same oral capsule?

**2c** 2. A patient is receiving his antihypertensive medication as 500 mg every 8 hours in the form of an oral capsule. The bioavailability of the drug from this capsule is 80% and the total body clearance in this patient is 2.5 L/hr. c. What is the $Cl_T$ of the drug in this patient while taking the 1000 mg orally every 8 hours?

A. 2.0 L/hr, the clearance times F
B. 2.5 L/hr, unchanged
C. 1.25 L/hr, half the clearance
D. 5 L/hr, twice the clearance

**2d** 2. A patient is receiving his antihypertensive medication as 500 mg every 8 hours in the form of an oral capsule. The bioavailability of the drug from this capsule is 80% and the total body clearance in this patient is 2.5 L/hr. d. What will be the average steady-state plasma concentration if the patient were taking 500 mg every 6 hours of the same oral capsule?

**2e** 2. A patient is receiving his antihypertensive medication as 500 mg every 8 hours in the form of an oral capsule. The bioavailability of the drug from this capsule is 80% and the total body clearance in this patient is 2.5 L/hr. e. What oral dose is required to reach an average steady-state plasma concentration of 30 mg/L from this drug product in this patient taken every 8 hours?

**2f** 2. A patient is receiving his antihypertensive medication as 500 mg every 8 hours in the form of an oral capsule. The bioavailability of the drug from this capsule is 80% and the total body clearance in this patient is 2.5 L/hr. f. Because the patient had problems swallowing the oral capsules he was shifted to 500 mg IV every 8 hours. What will be the average steady-state concentration of this drug in this patient?

# Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

## 1a

Her key

**Answer:** 10.46 mg/L

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **$D_0$** | 200-mg | $R = \frac{200}{2} = 100$ mg/hr |
| **infusion time** | 2 hours | the denominator of R and the t in $(1 - e^{-kt})$ |
| **second start** | Eight hours after the start | *not needed: only the first infusion is asked* |
| **t½** | 4 hours | gives $k = \frac{0.693}{4} = 0.17325$ hr$^{-1}$ |
| **$Cl_T$** | 2.8 L/hr | the denominator of the plateau $\frac{R}{Cl_T}$ |

### THE WORKING, ONE LINE AT A TIME

UNIT

$$R = \frac{200 \text{ mg}}{2 \text{ hr}} = 100 \text{ mg/hr}; \; k = \frac{0.693}{4 \text{ hr}} = 0.17325 \text{ hr}^{-1}$$

The infusion rate is the dose over the infusion time, 200 mg in 2 hours, and the rate constant is 0.693 over the 4-hour half-life; both are needed before the end-of-infusion equation can be used.

SETUP

$$C_p = \frac{R}{Cl_T}(1 - e^{-kt}), \text{ with } t = 2 \text{ hr}$$

Clearance is given directly, so R over $Cl_T$ is the plateau a continuous infusion would reach, and $(1 - e^{-kt})$ is the fraction of it reached after 2 hours.

ALGEBRA

$$C_p = \frac{100 \text{ mg/hr}}{2.8 \text{ L/hr}}(1 - e^{-(2)(0.17325)}) = (35.71 \text{ mg/L})(0.2929) = 10.46 \text{ mg/L}$$

mg/hr over L/hr leaves mg/L, because the hours cancel; two hours is half a half-life, so only 29% of the 35.71 mg/L plateau a continuous infusion would reach is attained when this infusion stops.

> **DOES THE NUMBER MAKE SENSE?**
> The plateau R over $Cl_T$ is 100 over 2.8, about 35.71 mg/L; two hours is half a half-life of 4 hours, so less than half of the plateau is reached (0.2929 of it), and 10.46 mg/L fits, below 35.71.

## 1b

Her key

**Answer:** 6.54 mg/L

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **$D_0$** | 200-mg | each infusion ends at 10.46 mg/L, as in part a |
| **infusion time** | 2 hours | sets the end times, 2 and 10 hr |
| **second start** | Eight hours after the start | the second infusion runs from 8 to 10 hr |
| **t½** | 4 hours | gives $k = \frac{0.693}{4} = 0.17325\ \text{hr}^{-1}$ |
| **$Cl_T$** | 2.8 L/hr | gave the end-of-infusion 10.46 mg/L in part a |
| **t** | 4 hours after the cessation | the second has declined 4 hr, the first 12 hr |

### THE WORKING, ONE LINE AT A TIME

SETUP

Time line: 0, 2, 8, 10, 14 hr

The first infusion runs from 0 to 2 hours; the second starts at 8 and ends at 10 hours; 4 hours after its end is 14 hours. The first infusion has then been declining for 12 hours and the second for 4.

SETUP

$C_p = (10.46\ \text{mg/L})\ e^{-(0.17325)(12)} + (10.46\ \text{mg/L})\ e^{-(0.17325)(4)}$

Each infusion contributes its end-of-infusion concentration declined for the time since its own end; the two add because elimination is first order.

ALGEBRA

$C_p = (10.46)(0.125) + (10.46)(0.5) = 1.31 + 5.23 = 6.54\ \text{mg/L}$

Twelve hours is three half-lives, leaving one eighth of the first infusion; four hours is one half-life, leaving half of the second.

> **DOES THE NUMBER MAKE SENSE?**
> Four hours is one half-life, so half of the second infusion remains, 10.46 × 0.5 = 5.23 mg/L; twelve hours is three half-lives, leaving an eighth of the first, 1.31; the sum, 6.54 mg/L, lies below 10.46.

## 2a

Her key

**Answer:** 20 mg/L

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **$D_0$** | 500 mg | the dose in the numerator |
| **τ** | every 8 hours | multiplies $Cl_T$: 2.5 × 8 = 20 L cleared per interval |
| **route** | oral capsule | F applies: only the absorbed part enters |
| **F** | 80% | F = 0.8 in the numerator |
| **$Cl_T$** | 2.5 L/hr | the denominator, with τ |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{avg}^{\infty} = \frac{FD_0}{Cl_T\,\tau}$$

Clearance is given directly, so the average at steady state is the absorbed dose per interval over the volume of plasma cleared per interval; F = 0.8 because only 80% of the capsule is absorbed.

ALGEBRA

$$C_{avg}^{\infty} = \frac{(0.8)(500\text{ mg})}{(2.5\text{ L/hr})(8\text{ hr})} = \frac{400\text{ mg}}{20\text{ L}} = 20\text{ mg/L}$$

L/hr × hr leaves litres, so mg over L is mg/L: 400 mg absorbed in each interval, spread over the 20 L cleared in it.

**DOES THE NUMBER MAKE SENSE?**

400 mg is absorbed every 8 hours and 2.5 L/hr × 8 hr = 20 L of plasma is cleared in that time, so 400 over 20 = 20 mg/L.

## 2b Her key

**Answer:** 40 mg/L

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **$D_0$** | 500 mg | *not needed: replaced by 1000 mg* |
| **τ** | every 8 hours | multiplies $Cl_T$: 2.5 × 8 = 20 L cleared per interval |
| **route** | oral capsule | F applies: only the absorbed part enters |
| **F** | 80% | F = 0.8 in the numerator |
| **$Cl_T$** | 2.5 L/hr | the denominator, with τ |
| **$D_0$ new** | 1000 mg | the dose in the numerator, twice the 500 mg |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{avg}^{\infty} = \frac{FD_0}{Cl_T\,\tau}, \text{ with } D_0 = 1000\text{ mg}$$

Only the dose changes; F, the clearance and the interval are the same as in part a, so the average scales with the dose, and the same equation is used with 1000 mg in place of 500 mg.

ALGEBRA

$$C_{avg}^{\infty} = \frac{(0.8)(1000\text{ mg})}{(2.5\text{ L/hr})(8\text{ hr})} = \frac{800\text{ mg}}{20\text{ L}} = 40\text{ mg/L}$$

Twice the dose gives twice the average, 40 mg/L against 20 mg/L, because the dose sits in the numerator and clearance does not change with the dose.

**DOES THE NUMBER MAKE SENSE?**

Twice the 500 mg dose gives twice the part a average of 20 mg/L: 40 mg/L; 800 mg absorbed over the 20 L cleared per interval.

## 2c Her key

**Answer:** B. 2.5 L/hr, unchanged

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **$D_0$** | 500 mg | the part a dose: rate in 50 mg/hr, $C_{avg}^{\infty}$ 20 mg/L |
| **τ** | every 8 hours | the denominator of the rate in and of $C_{avg}^{\infty}$ |
| **route** | oral capsule | F applies to the dose |
| **F** | 80% | F = 0.8 scales each dose |
| **$Cl_T$** | 2.5 L/hr | the patient value; the ratio returns it at both doses |
| **$D_0$ new** | 1000 mg | rate in 100 mg/hr, $C_{avg}^{\infty}$ 40 mg/L: both doubled |

### THE WORKING, ONE LINE AT A TIME

SETUP

$Cl_T = \frac{\text{rate of elimination}}{C_p}$; at steady state the rate of elimination equals the rate in, $\frac{F\,D_0}{\tau}$

Clearance is the elimination rate over the concentration. At steady state the body eliminates what is absorbed in each interval, so the rate of elimination is the absorbed dose over the interval.

ALGEBRA

At 500 mg: rate in $= \frac{(0.8)(500\text{ mg})}{8\text{ hr}} = 50$ mg/hr; $C_{avg}^{\infty} = \frac{(0.8)(500\text{ mg})}{(2.5\text{ L/hr})(8\text{ hr})} =$ 20 mg/L; $Cl_T = \frac{50\text{ mg/hr}}{20\text{ mg/L}} = 2.5$ L/hr

At 500 mg the rate in is 50 mg/hr and the average concentration, from part a, is 20 mg/L; their ratio returns the given clearance, 2.5 L/hr.

ALGEBRA

At 1000 mg: rate in $= \frac{(0.8)(1000\text{ mg})}{8\text{ hr}} = 100$ mg/hr; $C_{avg}^{\infty} = \frac{(0.8)(1000\text{ mg})}{(2.5\text{ L/hr})(8\text{ hr})}$ $= 40$ mg/L; $Cl_T = \frac{100\text{ mg/hr}}{40\text{ mg/L}} = 2.5$ L/hr

Doubling the dose doubles the rate in, 50 to 100 mg/hr, and the average, 20 to 40 mg/L, so their ratio, the clearance, stays 2.5 L/hr.

SETUP

Compare: $Cl_T = 2.5$ L/hr at 500 mg and 2.5 L/hr at 1000 mg; unchanged

The dose moved the elimination rate and the concentration together and left their ratio unchanged: 2.5 L/hr, which is why $C_{avg}^{\infty}$ scaled from 20 to 40 mg/L.

> **DOES THE NUMBER MAKE SENSE?**
> Clearance is a ratio of elimination rate to concentration, and the dose doubles both, 50 to 100 mg/hr and 20 to 40 mg/L, so the ratio cannot move: 2.5 L/hr at both doses. Multiplying by F or by 2 confuses clearance with the rate or the input.

Total body clearance, the volume of plasma cleared of drug per hour, belongs to the patient and the drug, not the dose. Doubling the dose doubles the rate of elimination, $Cl_T \times C_p$, because $C_p$ doubles; $Cl_T$ stays 2.5 L/hr.

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## 2d Her key

**Answer:** 26.67 mg/L

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **$D_0$** | 500 mg | the dose in the numerator |
| **τ** | every 8 hours | *not needed: replaced by 6 hours* |
| **route** | oral capsule | F applies: only the absorbed part enters |
| **F** | 80% | F = 0.8 in the numerator |
| **$Cl_T$** | 2.5 L/hr | the denominator, with τ |
| **τ new** | every 6 hours | multiplies $Cl_T$: 2.5 × 6 = 15 L cleared per interval |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{avg}^{\infty} = \frac{FD_0}{Cl_T\,\tau}, \text{ with } \tau = 6 \text{ hr}$$

Only the interval changes; a shorter interval means the same 400 mg is absorbed over fewer hours, so less plasma is cleared per dose.

ALGEBRA

$$C_{avg}^{\infty} = \frac{(0.8)(500 \text{ mg})}{(2.5 \text{ L/hr})(6 \text{ hr})} = \frac{400 \text{ mg}}{15 \text{ L}} = 26.67 \text{ mg/L}$$

The average rises in the ratio 8 over 6: 20 mg/L × 8 over 6 = 26.67 mg/L, because τ sits in the denominator; the same 400 mg is now absorbed every 6 hours instead of every 8.

> **DOES THE NUMBER MAKE SENSE?**
> Only 15 L is cleared in 6 hours against 20 L in 8, so the average rises from the part a value of 20 mg/L by the ratio 8 to 6, to a little under 27 mg/L; 26.67 fits.

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## 2e Her key

**Answer:** 750 mg

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **$D_0$** | 500 mg | *not needed: the dose is asked for* |
| **τ** | every 8 hours | multiplies $Cl_T$: 2.5 × 8 = 20 L cleared per interval |
| **route** | oral capsule | F applies: only the absorbed part enters |
| **F** | 80% | divides: $D_0 = \frac{C_{avg}^{\infty}\, Cl_T\, \tau}{F}$ |
| **$Cl_T$** | 2.5 L/hr | the denominator, with τ |
| **$C_{avg}^{\infty}$** | 30 mg/L | the target, multiplied by $Cl_T$ and τ |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{avg}^{\infty} = \frac{FD_0}{Cl_T\,\tau}, \text{ so } D_0 = \frac{C_{avg}^{\infty}\,Cl_T\,\tau}{F}$$

The average equation is rearranged for the dose; dividing by F = 0.8 makes the oral dose larger than the amount that has to be absorbed, because only 80% of a capsule reaches the circulation.

ALGEBRA

$$D_0 = \frac{(30\text{ mg/L})(2.5\text{ L/hr})(8\text{ hr})}{0.8} = \frac{600\text{ mg}}{0.8} = 750\text{ mg}$$

mg/L × L/hr × hr leaves mg: 600 mg must be absorbed per interval, and 750 mg must be swallowed for 600 mg to be absorbed.

> **DOES THE NUMBER MAKE SENSE?**
> 500 mg gave 20 mg/L (part a), so 30 mg/L needs half as much again: 750 mg. The dose exceeds the 600 mg that must be absorbed because only 80% of a capsule is absorbed.

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## 2f Her key

**Answer:** 25 mg/L

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **$D_0$** | 500 mg | the dose in the numerator |
| **τ** | every 8 hours | multiplies $Cl_T$: 2.5 × 8 = 20 L cleared per interval |
| **route** | oral capsule | *not needed: replaced by the IV route* |
| **F** | 80% | *not needed: an IV dose has F = 1* |
| **$Cl_T$** | 2.5 L/hr | the denominator, with τ |
| **route new** | 500 mg IV | F = 1: the whole dose enters |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{avg}^{\infty} = \frac{FD_0}{Cl_T\,\tau}, \text{ with } F = 1$$

An IV dose enters the circulation whole, so F = 1 and the whole 500 mg counts; the clearance and the interval are the same as for the capsule, so only F changes.

ALGEBRA

$$C_{avg}^{\infty} = \frac{500\text{ mg}}{(2.5\text{ L/hr})(8\text{ hr})} = \frac{500\text{ mg}}{20\text{ L}} = 25\text{ mg/L}$$

L/hr × hr leaves litres, so mg over L is mg/L; the IV regimen gives 25 mg/L against 20 mg/L for the same dose by capsule, because the capsule delivered only 400 mg of each 500 mg.

> **DOES THE NUMBER MAKE SENSE?**

> By IV the whole 500 mg enters instead of 80% of it, so the average rises from the capsule value of 20 mg/L by a quarter: 25 mg/L, 500 mg over the 20 L cleared per interval.

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